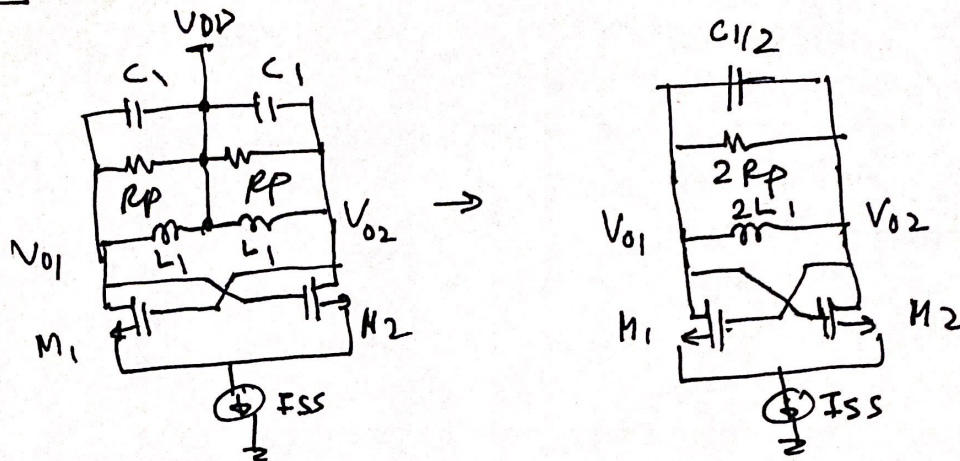


Homework 6:

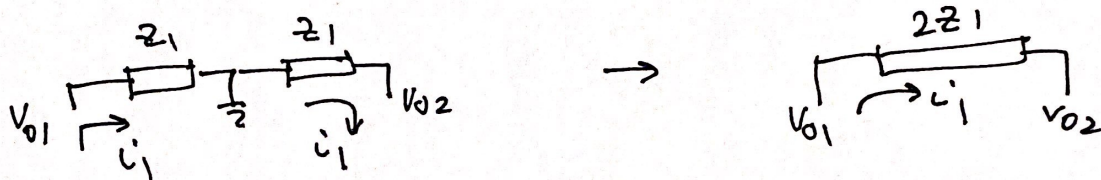
chapter 8, vcos:

1

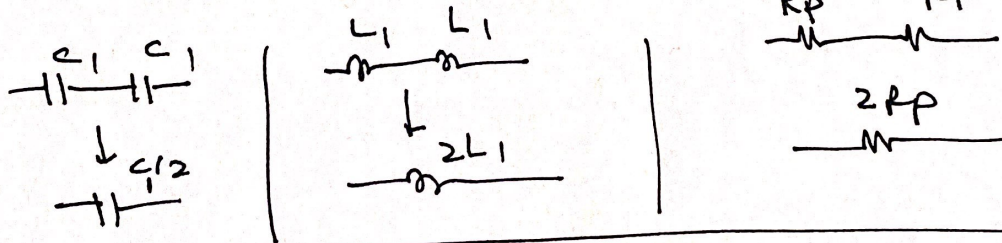
problem 8.7:



$V_{01} = -V_{02} \rightarrow$ The mid. point is ac ground



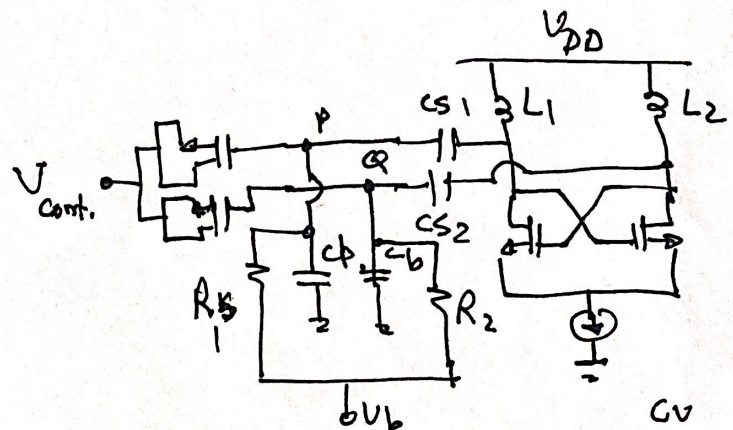
$$i_1 = \frac{V_{01} - V_{02}}{2Z_1} = \frac{2V_{01}}{2Z_1} = \frac{V_{01}}{Z_1} = \frac{-V_{02}}{Z_1}$$



problem 8.8:

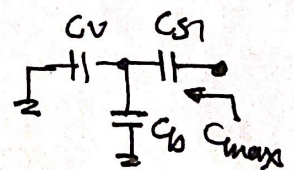
for varactor:

$$\begin{cases} V_{max} \rightarrow C_{max} \\ V_{min} \rightarrow C_{min} \end{cases}$$



$$f_{min} = \frac{1}{2\pi\sqrt{L_1(C_1 + C_{max})}}$$

$$f_{max} = \frac{1}{2\pi\sqrt{L_1(C_1 + C_{min})}}$$



If $C_{S1} \gg C_{Vmax} \gg C_b$, $C_{Vmin} > C_b$

$$C_{Vmax} \approx \frac{C_{S1} C_{Vmax}}{C_{S1} + C_{Vmax}}, \quad C_{Vmin} \approx C_{Vmin} + C_b \rightarrow \frac{C_{Vmax}}{C_{Vmin}} \approx \frac{C_{S1} C_{Vmax} (C_{S1} + C_b)}{C_{S1} + C_{Vmax}}$$

Assume $C_{S1} \approx 8 C_{Vmax}$, $C_{Vmax} = 4 C_{Vmin}$, $C_b \approx \frac{1}{2} C_{Vmin}$

$$C_{Vmax} \approx \frac{8}{9} C_{Vmax}, \quad C_{Vmin} = \frac{3}{2} C_{Vmin} = \frac{3}{8} C_{Vmax}$$

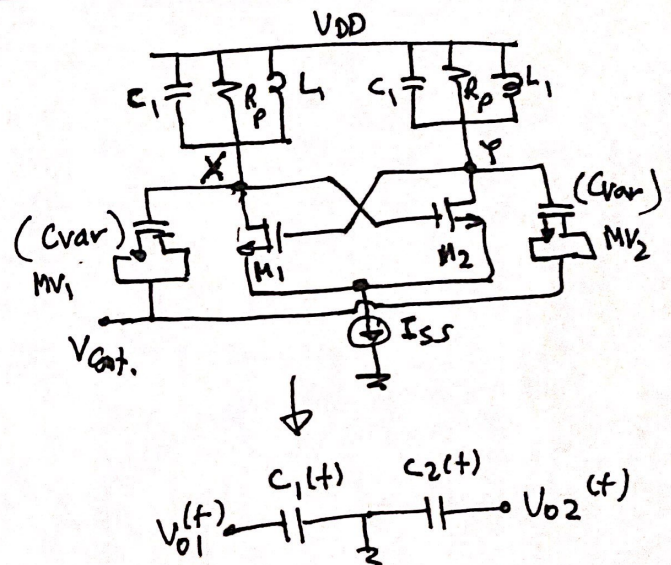
$$\rightarrow \frac{C_{Vmax}}{C_{Vmin}} = \frac{8/9}{3/8} \rightarrow \frac{C_{Vmax}}{C_{Vmin}} \approx 2.37$$

problem 8.13

- $C_{var} = C_0 (1 + \alpha_1 V + \alpha_2 V^2)$
- Complete current steering
- low frequency noise component of $I_n = I_m \cos(\omega_m t)$ in FSS

(a) AM noise resulting from I_n ;

$$V_{o1}(t) = -V_{o2}(t) = \underbrace{A(t)}_{\text{Modulation due to } I_n} \sin(\omega_0 t)$$



$$C_1(t) = C_0 \left[1 + \alpha_1 (V_{dc} + A(t) \sin(\omega_0 t)) + \alpha_2 (V_{dc} + A(t) \sin(\omega_0 t))^2 \right]$$

$$= C_0 (1 + \alpha_1 V_{dc} + \alpha_2 V_{dc}^2) + C_0 (\alpha_1 A(t) \sin(\omega_0 t) + 2\alpha_1 V_{dc} A(t) \sin(\omega_0 t) + \alpha_2 A^2(t) \sin^2(\omega_0 t))$$

$$C_2(t) = C_0 \left[1 + \alpha_1 (V_{dc} - A(t) \sin(\omega_0 t)) + \alpha_2 (V_{dc} - A(t) \sin(\omega_0 t))^2 \right]$$

$$= C_0 (1 + \alpha_1 V_{dc} + \alpha_2 V_{dc}^2) + C_0 (-\alpha_1 A(t) \sin(\omega_0 t) - 2\alpha_1 V_{dc} A(t) \sin(\omega_0 t) + \alpha_2 A^2(t) \sin^2(\omega_0 t))$$

(b) Average value of C_{var} vs. $A(t)$:

$$C_{avg}(t) = \frac{C_1(t) + C_2(t)}{2} = C_0(1 + \alpha_1 V_{DC} + \alpha_2 V_{DC}^2) + C_0 (\alpha_2 A^2(t) \sin^2(\omega_0 t))^{-3}$$

$$C_{avg}(t) = C_0(1 + \alpha_1 V_{DC} + \alpha_2 V_{DC}^2) + C_0 \left[\frac{\alpha_2}{2} (1 - \cos(2\omega_0 t)) A^2(t) \right]$$

$$\{ A(t) = V_{AM, SE} = \frac{2}{\pi} (I_{SS} + I_m \cos(\omega_m t)) \} R_p = \frac{2}{\pi} (I_{SS} + I_m \cos(\omega_m t)) R_p$$

$$\{ A^2(t) = \frac{4}{\pi^2} R_p^2 (I_{SS}^2 + 2 I_m I_{SS} \cos(\omega_m t) + \cancel{I_m^2 \cos^2(\omega_m t)}) \}$$

too small

~~*) Average value of C_{var} vs. $A(t)$:~~

$$\{ \Delta C_{var}(t) = C_0 \frac{\alpha_2}{2} A^2(t) \} = C_0 \frac{\alpha_2}{2} \cdot \frac{4}{\pi^2} R_p^2 (I_{SS}^2 + 2 I_{SS} I_m \cos(\omega_m t))$$

$$\Delta C_{var}(t) = \underbrace{C_0 \frac{\alpha_2}{2} \cdot \frac{4}{\pi^2} R_p^2 I_{SS}^2}_{\text{Static shift in frequency}} + C_0 \alpha_2 I_m I_{SS} \frac{4}{\pi^2} R_p^2 \cos(\omega_m t)$$

$$\Delta \omega_{osc}(t) \approx \frac{1}{\sqrt{L_1 C_1}} \frac{\Delta C_{var}(t)}{2 C_1}, \quad \omega_0 = \frac{1}{\sqrt{L_1 C_1}} \rightarrow \frac{\Delta \omega_{osc}}{\omega_{osc}} \approx \frac{\Delta C_{var}(t)}{2 C_1}$$

$$\rightarrow \frac{\Delta \omega_{osc}(t)}{\omega_{osc}} \approx \frac{C_0 \alpha_2 I_m I_{SS} \frac{4}{\pi^2} R_p^2 \cos(\omega_m t)}{2 C_1} \quad (\omega_{osc} = \omega_0)$$

$$(c) \quad \Delta \phi(t) = \int \Delta \omega_{osc}(t) dt = \int \left(\frac{\alpha_2 C_0}{2 C_1} \right) \left(\frac{4}{\pi^2} R_p^2 \right) (I_m I_{SS}) \omega_0 \cos(\omega_m t) dt$$

$$\Delta \phi(t) = + \underbrace{\left(\frac{\alpha_2 C_0}{2 C_1} \right) \left(\frac{\omega_0}{\omega_m} \right) \left(\frac{4}{\pi^2} R_p^2 \right) (I_{SS})}_{\Phi_0} \underbrace{(I_m \sin(\omega_m t))}_{\Phi_0 = + \left(\frac{\alpha_2 C_0}{2 C_1} \right) \left(\frac{\omega_0}{\omega_m} \right) \left(\frac{4 R_p^2}{\pi^2} \right) (I_{SS} I_m)} = \Phi_0 \sin(\omega_m t)$$

spur level: $V_o = A_0 \cos(\omega_0 t + \phi_0 \sin(\omega_m t))$

$$V_i \approx A_0 [\cos(\omega_0 t) - \phi_0 \sin(\omega_m t) \sin(\omega_0 t)]$$

$$V_o \approx A_0 \cos(\omega_0 t) - \frac{A_0 \phi_0}{2} [\cos(\omega_0 - \omega_m)t - \cos(\omega_0 + \omega_m)t]$$

$\rightarrow$ each spur level $\frac{A_0}{2} \phi_0 \times \frac{1}{A_0} = \frac{\phi_0}{2}$ (relative to fundamental)

phase noise gain $\rightarrow$

$$\left| \frac{\dot{\phi}_0}{I_{SS} I_m} \right| = \left(\frac{\alpha_2 C_0}{2 C_1} \right) \left(\frac{4 R_P^2}{\pi^2} \right) \left(\frac{\omega_0}{\omega_m} \right) I_{SS}$$

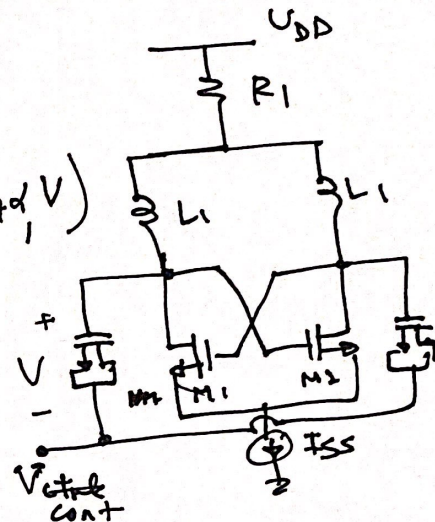
$$V_0 = \frac{2}{\pi} R_P I_{SS}$$

$$\left| \frac{\dot{\phi}_0}{I_{SS} I_m} \right| = \left(\frac{\alpha_2 C_0}{2 C_1} \right) V_0^2 \left(\frac{\omega_0}{\omega_m} \right) \cdot \frac{1}{I_{SS}}$$

problem 8.17:

- VCO with level-shift resistor
- $V_{cont.}$ is given
- parasitics are neglected

$$C_{var} = C_0 (1 + \alpha_1 V)$$



(a) $V = V_{GS, var} \approx V_{DD} - I_{SS} (R_1) - V_{cont.}$

$$C_{var} = C_0 [1 + \alpha_1 (V_{DD} - I_{SS} R_1 - V_{cont.})]$$

$$\omega_0 = \frac{1}{\sqrt{L_1 C_0 [1 + \alpha_1 (V_{DD} - R_1 I_{SS} - V_{cont.})]}}$$

$$\gamma = \frac{\partial \omega_0}{\partial I_{SS}} = \frac{1}{\sqrt{L_1 C_0}} \frac{(-1/2 \alpha_1 R_1)}{[1 + \alpha_1 (V_{DD} - R_1 I_{SS} - V_{cont.})]^{3/2}} = \frac{\frac{\omega_0}{2} \alpha_1 R_1}{[1 + \alpha_1 (V_{DD} - R_1 I_{SS} - V_{cont.})]^{3/2}}$$

(b) Noise Component, $\Delta I_{SS}^{(t)} = I_m \cos(\omega_m t)$

$$\omega^{(t)} = \omega_0 + \left(\frac{\partial \omega_0}{\partial I_{SS}} \right) \Delta I_{SS}^{(t)} \rightarrow \text{Taylor expansion}$$

$$\omega = \omega_0 + \gamma I_m \cos(\omega_m t)$$

$$\phi(t) = \int \omega(t) dt = \omega_0 t + \frac{\gamma I_m}{\omega_m} \sin(\omega_m t)$$

$$y(t) = A_0 \cos \left[\omega_0 t + \left(\frac{\gamma I_m}{\omega_m} \right) \sin(\omega_m t) \right]$$

$$\approx A_0 \cos(\omega_0 t) - A_0 \left(\frac{\gamma I_m}{\omega_m} \right) \sin(\omega_m t) \sin(\omega_0 t)$$

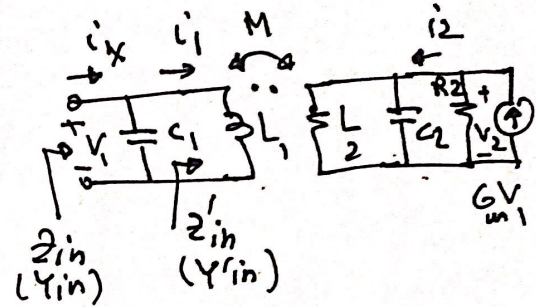
#5

$$y(t) \approx A_0 \cos(\omega_0 t) - \frac{A_0}{2} \left(\frac{\delta I_m}{\omega_m} \right) \left[\cos(\omega_0 - \omega_m)t - \cos(\omega_0 + \omega_m)t \right]$$

Relative magnitude of spurs = $\frac{\delta I_m}{\omega_m}$

problem 8.18

- Simplified model of 'dual-mode' oscillator.
- Voltage-dependent Current Source models transistors
- Circuit oscillators if $Z_{in} \rightarrow \infty$ for $s = j\omega$



(a) $Z_{in} = ?$

$$\begin{cases} V_1 = (sL_1) i_1 + (sM) i_2 \\ V_2 = (sL_2) i_2 + (sM) i_1 \end{cases}$$

$$G_m V_1 = V_2 \left[sC_2 + \frac{1}{R_2} \right] + i_2$$

$$G_m (i_1 (sL_1) + i_2 (sM)) = i_2 + (i_2 (sL_2) + i_1 (sM)) \left(sC_2 + \frac{1}{R_2} \right)$$

$$i_2 \left[1 + sL_2 \left(sC_2 + \frac{1}{R_2} \right) - G_m sM \right] = i_1 \left[G_m sL_1 - sM \left(sC_2 + \frac{1}{R_2} \right) \right]$$

$$\frac{i_1}{i_2} = \frac{1 + s^2 L_2 C_2 + \frac{sL_2}{R_2} - G_m sM}{s(G_m L_1 - \frac{M}{R_2}) - s^2 M C_2}$$

$$\begin{cases} i_x = V_1 C_1 s + i_1 \Rightarrow Z_{in} = \frac{1}{sC_1} \parallel Z'_{in}, \quad Z'_{in} = \frac{V_1}{i_1} \\ Z_{in} = \frac{V_1}{i_x} \Rightarrow Y_{in} = \frac{i_x}{V_1} = \frac{i_1 + V_1 C_1 s}{V_1} = C_1 s + \frac{i_1}{V_1} = C_1 s + Y'_{in} \end{cases}$$

$$Z'_{in} = \frac{(sL_1) i_1 + (sM) i_2}{i_1} = sL_1 + (sM) \left(\frac{i_2}{i_1} \right)$$

$$Z'_{in} = sL_1 + \left[\frac{(sM) \left[s(G_m L_1 - \frac{M}{R_2}) - s^2 M C_2 \right]}{(1 + s^2 L_2 C_2) + \left(\frac{sL_2}{R_2} - G_m sM \right)} \right]$$

$$Z'_{in} = \frac{SL_1 + S^3 L_1 L_2 C_2 - S^3 M^2 C_2 + \frac{S^2 L_1 L_2}{R_2} - \cancel{G_m L_1 M S^2} + \cancel{S^2 M G_m L_1} - \frac{S M^2}{R_2}}{(1 + S^2 L_2 C_2) + (\frac{S L_2}{R_2} - G_m S M)} \quad 6$$

$$Z'_{in} = \frac{SL_1 + S^3 C_2 (L_1 L_2 - M^2) + \frac{S^2}{R_2} (L_1 L_2 - M^2)}{(1 + S^2 L_2 C_2) + (\frac{S L_2}{R_2} - G_m S M)} \quad (K^2 = \frac{M^2}{L_1 L_2})$$

$$Z'_{in} = SL_1 \cdot \frac{1 + S^2 L_2 (1 - K^2) C_2 + \frac{S L_2}{R_2} (1 - K^2)}{(1 + S^2 L_2 C_2) + (\frac{S L_2}{R_2} - G_m S M)}$$

$$Y'_{in} = \frac{1}{Z'_{in}}, \quad Y_{in} = G_S + Y'_{in} = C_1 S + \frac{1}{SL_1} \frac{(1 + S^2 L_2 C_2) + (\frac{S L_2}{R_2} - S M G_m)}{1 + S^2 L_2 C_2 (1 - K^2) + \frac{S L_2}{R_2} (1 - K^2)}$$

$$Y_{in} = \frac{S^2 L_1 C_1 \left[1 + S^2 L_2 C_2 (1 - K^2) + \frac{S L_2}{R_2} (1 - K^2) \right] + (1 + S^2 L_2 C_2) + (\frac{S L_2}{R_2} - S M G_m)}{(S L_1) (1 + S^2 L_2 C_2 (1 - K^2) + \frac{S L_2}{R_2} (1 - K^2))}$$

$$(b) \text{ If } Z_{in} \rightarrow \infty \Rightarrow Y_{in} \rightarrow 0 \rightarrow Y_{in}(s) = \frac{N(s)}{D(s)} = 0 \rightarrow \underline{D(s) = 0}$$

$$0 = S^2 L_1 C_1 + S^4 L_1 C_1 L_2 C_2 (1 - K^2) + \frac{S^3 L_1 L_2 C_1}{R_2} (1 - K^2) + (1 + S^2 L_2 C_2) + (\frac{S L_2}{R_2} - S M G_m)$$

$$\text{if } S = j\omega \rightarrow -\omega^2 L_1 C_1 + \omega^4 L_1 C_1 L_2 C_2 (1 - K^2) - \frac{j\omega^3 L_1 L_2 C_1}{R_2} (1 - K^2) + 1 - \omega^2 L_2 C_2 + \frac{j\omega L_2}{R_2} - j\omega M G_m = 0$$

$$= 0 \quad (1)$$

$$\begin{cases} \text{Re}(D(j\omega)) = 0 \rightarrow -\omega^2 L_1 C_1 + \omega^4 L_1 C_1 L_2 C_2 (1 - K^2) + 1 - \omega^2 L_2 C_2 = 0 & (1) \\ \text{Im}(D(j\omega)) = 0 \rightarrow -\frac{\omega^3 L_1 L_2 C_1}{R_2} (1 - K^2) + \frac{\omega L_2}{R_2} - \omega M G_m = 0 & (2) \end{cases}$$

$$(1) : \left[\omega^4 L_1 C_1 L_2 C_2 (1 - K^2) - \omega^2 (L_2 C_2 + L_1 C_1) + 1 = 0 \right]$$

$$(2) : \left[G_m = \frac{L_2}{R_2} \frac{1}{M} \left[1 - \omega^2 L_1 C_2 (1 - K^2) \right] \right]$$

$$\text{For } L_1 = L_2 = L, \quad C_1 = C_2 = C, \quad M = LK \quad (M < L)$$

$$\omega^2 = \frac{LC \pm LC \sqrt{1 - (1 - K^2)}}{L^2 C^2 (1 - K^2)} \rightarrow$$

$$\omega^2 = \frac{1}{LC(1 \mp K)}$$

$$\text{or } \left[\omega^2 = \frac{1}{(L+M)C}, \frac{1}{(L-M)C} \right] \rightarrow \left[\omega_1 = \frac{1}{\sqrt{(L+M)C}}, \omega_2 = \frac{1}{\sqrt{(L-M)C}} \right]$$

and $G_m = \frac{L}{R_2 M} \left[1 - \frac{LC(1-K^2)}{LC(1 \mp K)} \right]$

7

$$G_m = \frac{L}{R_2 M} [1 - (1 \pm K)] = \mp \frac{KL}{R_2 M} = \mp \frac{1}{R_2}$$

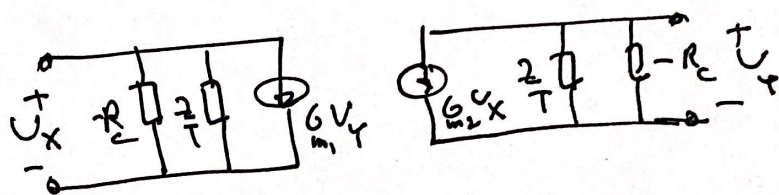
for $\omega_1 = \frac{1}{\sqrt{(L+M)C}} \rightarrow G_m R_2 = 1$
 for $\omega_2 = \frac{1}{\sqrt{(L-M)C}} \rightarrow G_m R_2 = -1$

Additional problems for QVCOs:

problem 8.15:

$Z_T = (L_1 S) \parallel (C_1 S)^{-1} \parallel R_p$

Start-up condition?



$$\begin{cases} V_x = -G_{m1} (Z_T \parallel (-R_c)) V_y \\ V_y = -G_{m2} (Z_T \parallel (-R_c)) V_x \end{cases} \rightarrow \begin{cases} G_{m1} V_y^2 - G_{m2} V_x^2 = 0 \\ G_{m1} G_{m2} [Z_T \parallel (-R_c)]^2 = 1 \end{cases}$$

Case (1) if $G_{m1} = G_{m2} \rightarrow V_y^2 = V_x^2$ or $V_y = \pm V_x$

$$G_m^2 \left[\frac{-Z_T R_c}{Z_T - R_c} \right]^2 = 1 \quad \text{or} \quad G_m \left[\frac{-Z_T R_c}{Z_T - R_c} \right] = \pm 1$$

$$\begin{cases} G_m \left(\frac{-Z_T R_c}{Z_T - R_c} \right) = -1 \quad (V_x = V_y) \\ G_m \left(\frac{-Z_T R_c}{Z_T - R_c} \right) = +1 \quad (V_x = -V_y) \end{cases} \rightarrow \begin{cases} Z_T(1 - G_m R_c) = R_c \\ Z_T(1 + G_m R_c) = R_c \end{cases} \rightarrow \begin{cases} Z_T \left(\frac{1}{R_c} - G_m \right) = 1 \\ Z_T \left(\frac{1}{R_c} + G_m \right) = 1 \end{cases}$$

@ resonance $Z_T = R_p \rightarrow$
 $V_x = V_y \rightarrow R_p \left[\frac{1}{R_c} - G_m \right] = 1$
 $V_x = -V_y \rightarrow R_p \left[\frac{1}{R_c} + G_m \right] = 1$

for $V_x = V_y \rightarrow G_m$ should be negative to make the start-up easier
 for $V_x = -V_y \rightarrow G_m$ should be positive to make the start-up easier

Case (2) if $G_{m1} = -G_{m2} \rightarrow v_y^2 = -v_x^2 \rightarrow v_y = \pm j v_x$ #8

$$-G_m^2 \left[\frac{-Z_T R_C}{Z_T - R_C} \right]^2 = 1 \rightarrow j G_m \left[\frac{-Z_T R_C}{Z_T - R_C} \right]^2 = \pm 1$$

$$\begin{cases} v_y = j v_x \rightarrow \left(\frac{1}{R_C} - j G_m \right) Z_T = 1 \\ v_y = -j v_x \rightarrow \left(\frac{1}{R_C} + j G_m \right) Z_T = 1 \end{cases}$$

$$Z_T = \frac{1}{s C_1 + \frac{1}{L_1 s} + R_P}, \quad s = j\omega \rightarrow Z_T(j\omega) = \frac{1}{j\omega C_1 \left[1 - \frac{1}{\omega^2 L_1 C_1} \right] + \frac{1}{R_P}}$$

For $\omega = \omega_0 + \Delta\omega$, ($\Delta\omega \ll \omega_0$) $\rightarrow Z_T(j\omega) \approx \frac{1}{2j\omega_0 C_1 \Delta\omega + \frac{1}{R_P}} = \frac{R_P}{1 + j 2Q \left(\frac{\Delta\omega}{\omega_0} \right)}$

$$Q = \omega_0 C_1 R_P$$

Case (2)
($v_y = j v_x$) $\rightarrow \left(\frac{1}{R_C} - j G_m \right) \left(\frac{R_P}{1 + j 2Q \left(\frac{\Delta\omega}{\omega_0} \right)} \right) = 1 \rightarrow \frac{R_P}{R_C} - j G_m R_P = 1 + j 2Q \left(\frac{\Delta\omega}{\omega_0} \right)$

$$\left[\frac{R_P}{R_C} \approx 1 \right]$$

and $-G_m R_P = 2Q \left(\frac{\Delta\omega}{\omega_0} \right)$ or $\left[\frac{\Delta\omega}{\omega_0} = -\frac{G_m R_C}{2Q} \right]$

($v_y = -j v_x$) $\rightarrow \left(\frac{1}{R_C} + j G_m \right) \left(\frac{R_P}{1 + j 2Q \left(\frac{\Delta\omega}{\omega_0} \right)} \right) = 1 \rightarrow \frac{R_P}{R_C} + j G_m R_P = 1 + j 2Q \left(\frac{\Delta\omega}{\omega_0} \right)$

$$\left[\frac{R_P}{R_C} = 1 \right]$$

and $G_m R_P = 2Q \left(\frac{\Delta\omega}{\omega_0} \right) \rightarrow \left[\frac{\Delta\omega}{\omega_0} = \frac{G_m R_C}{2Q} \right]$

Case (1)

($v_y = v_x$) $\rightarrow \left(\frac{1}{R_C} - G_m \right) \left(\frac{R_P}{1 + j 2Q \left(\frac{\Delta\omega}{\omega_0} \right)} \right) = 1 \rightarrow \frac{R_P \left(\frac{1}{R_C} - G_m \right)}{1 + j 2Q \left(\frac{\Delta\omega}{\omega_0} \right)}$

$$\rightarrow \left[R_P \left[\frac{1}{R_C} - G_m \right] = 1 \right]$$

$\frac{\Delta\omega}{\omega_0} = 0$!
Not accurate

($v_y = -v_x$) $\rightarrow \left(\frac{1}{R_C} + G_m \right) \left(\frac{R_P}{1 + j 2Q \left(\frac{\Delta\omega}{\omega_0} \right)} \right) = 1$

$$\rightarrow \left[R_P \left[\frac{1}{R_C} + G_m \right] = 1 \right]$$

$$\rightarrow \frac{R_P \left(\frac{1}{R_C} + G_m \right)}{1 + j 2Q \left(\frac{\Delta\omega}{\omega_0} \right)}$$

$\frac{\Delta\omega}{\omega_0} = 0$! Not accurate

problem 8.16

9

$$\Delta\omega = \frac{\omega_0}{2Q_T} \tan^{-1} \left(\frac{I_{D3}}{I_{D1}} \right)$$

$$\rightarrow \text{for small-signal operation} \rightarrow \Delta\omega = \frac{\omega_0}{2Q_T} \tan^{-1} \left(\frac{g_{m3}}{g_{m1}} \right)$$

→ The tail current of the coupling transistors, I_{T1} contains a flicker noise component ($I_n \ll I_{T1}$)

$$g'_{m3} = \sqrt{2\mu_n C_{ox} \left(\frac{W}{L}\right)_3 \frac{(I_{T1} + I_n)}{2}}$$

(adding the noise)

$$\left[\begin{aligned} \Delta\omega &\stackrel{?}{=} f(I_n) \text{ (Linear)} \\ K_{VCO} &= \frac{\partial(\Delta\omega)}{\partial I_n} \end{aligned} \right. ?$$

$$g'_{m3} \approx \sqrt{2\mu_n C_{ox} \left(\frac{W}{L}\right)_3 \frac{I_{T1}}{2}} \left(1 + \frac{I_n}{2I_{T1}} \right)$$

$$\rightarrow \Delta\omega = \frac{\omega_0}{2Q_T} \tan^{-1} \left[\frac{\sqrt{2\mu_n C_{ox} \left(\frac{W}{L}\right)_3 \frac{I_{T1}}{2}} \left(1 + \frac{I_n}{2I_{T1}} \right)}{g_{m1}} \right]$$

$$\Delta\omega = \frac{\omega_0}{2Q_T} \tan^{-1} \left[\frac{g_{m3}}{g_{m1}} \left(1 + \frac{I_n}{2I_{T1}} \right) \right], \quad \frac{g_{m3}}{g_{m1}} \left(1 + \frac{I_n}{2I_{T1}} \right) =$$

$$\tan \left[\underbrace{\left(\frac{\Delta\omega}{\omega_0} \right) (2Q_T)}_x \right] \rightarrow \frac{g_{m3}}{g_{m1}} \left(1 + \frac{I_n}{2I_{T1}} \right) = \left(\frac{\Delta\omega}{\omega_0} \right) (2Q_T)$$

$$\left[\begin{aligned} \tan(x) &\sim x \quad (\Delta\omega \ll \omega_0) \\ \text{if } x &\text{ small} \end{aligned} \right.$$

$$\Delta\omega = \left(\frac{\omega_0}{2Q_T} \right) \left(\frac{g_{m3}}{g_{m1}} \right) \left(1 + \frac{I_n}{2I_{T1}} \right)$$

$$\frac{\partial(\Delta\omega)}{\partial I_n} = \left(\frac{\omega_0}{2Q_T} \right) \left(\frac{g_{m3}}{g_{m1}} \right) \left(\frac{1}{2I_{T1}} \right), \quad \left[K_{VCO} = \left(\frac{\omega_0}{2Q_T} \right) \left(\frac{g_{m3}}{g_{m1}} \right) \left(\frac{1}{2I_{T1}} \right) \right]$$